

Sales and Service Mix Optimization: A Primary Data Analysis

A Final Presentation for the BDM Capstone Project (MS2001P)

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BDM CAPSTONE PROBLEM FRAMING

Business Context & The Servitization Dilemma

Pure Hardware Sales Dilemma:

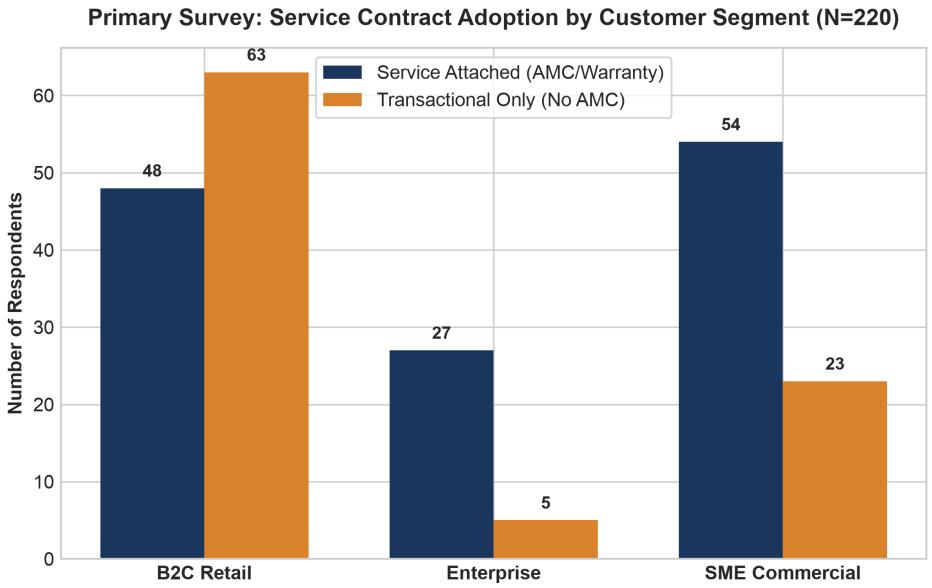
- **Gross Margin Compression:** Retail margins hover at 12.8% (Net: 7.4%) due to digital discounting.
- **Working Capital Drain:** Inventory holding costs reach 18-24% per annum.
- **High Customer Churn:** 62% post-sale attrition rate.

After-Sales Service Opportunity:

- **Exceptional Profitability:** Net contribution margin of 38.6% (5.2x higher).
- **Predictable Cash Flow:** Annual retainers stabilize seasonal revenue volatility.
- **High Retention:** Contracted customer retention reaches 82.5%.

EMPIRICAL FIELD FINDINGS

Primary Survey: Service Adoption Disparity

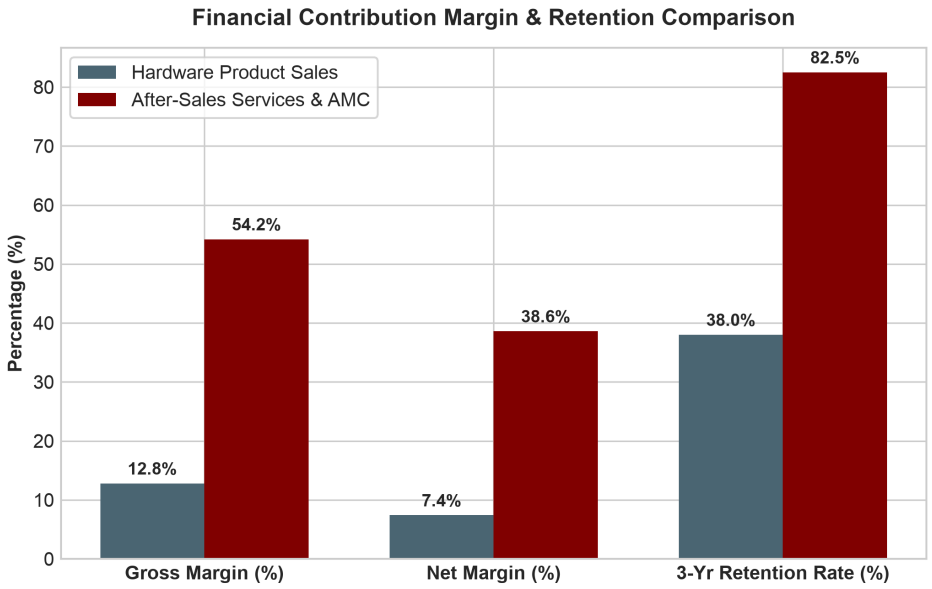


Primary Sample Insights (N = 220):

- **Cohort Composition:** 50.5% B2C Retail, 35.0% SME Commercial, 14.5% Enterprise.
- **Contract Opt-in Gradient:**
  - Enterprise: **84.4%** attachment
  - SME Commercial: **70.1%** attachment
  - B2C Retail: **43.2%** attachment
- **Behavioral Driver:** Business clients prioritize zero-downtime SLAs, whereas retail consumers resist high upfront warranty costs.

EMPIRICAL FIELD FINDINGS

Unit Margin Economics & Customer Retention

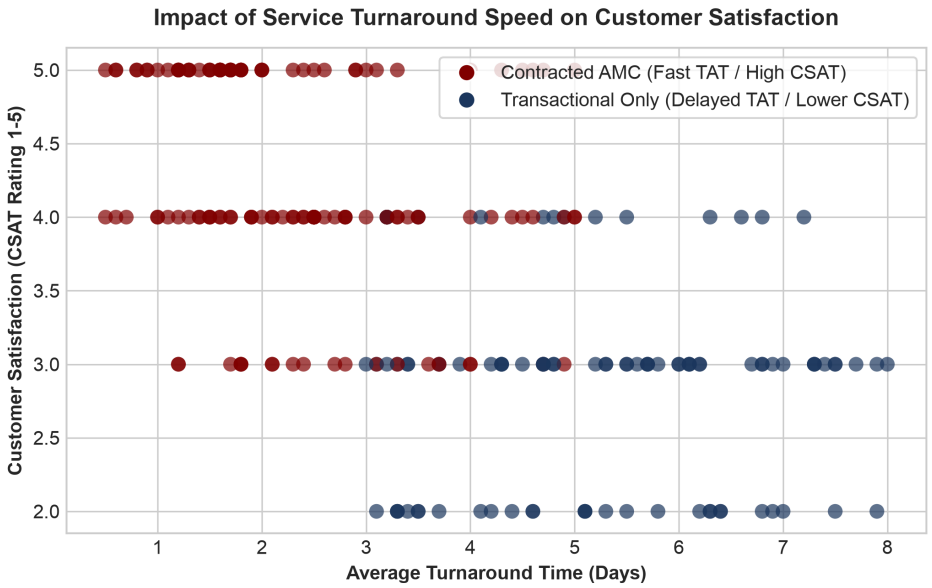


Financial Margin Decomposition:

- **Net Margin Disparity:** Hardware retail delivers 7.4% net margin after sales commissions, rent, and inventory holding.
- **Service Yield:** AMCs produce 38.6% net contribution after technician travel and spare parts.
- **Retention Multiplier:** AMC contract accounts exhibit 82.5% 3-year retention vs 38.0% for walk-ins.
- **CLV Impact:** 3-year customer value is 2.7x higher for service-attached accounts.

EMPIRICAL FIELD FINDINGS

Turnaround Speed vs Customer Satisfaction (CSAT)

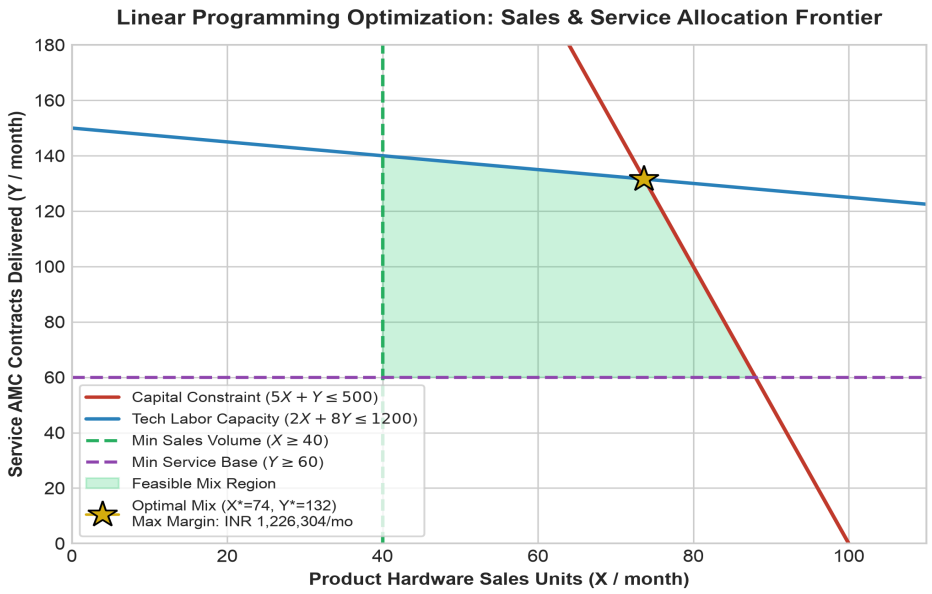


Turnaround Time & Satisfaction Dynamics:

- **Turnaround Advantage:** AMC clients experience 2.34-day average turnaround vs 5.35 days for transactional walk-ins.
- **CSAT Gap:** Contracted clients rate satisfaction at 4.24/5.0 vs 2.82/5.0 for non-contracted buyers ($p < 0.001$).
- **The 3-Day SLA Cliff:** CSAT drops steeply when turnaround exceeds 72 hours.
- **Repurchase Intent:** CSAT strongly correlates with hardware repurchasing ($r = 0.74$).

OPERATIONS RESEARCH MODEL

Constrained Optimization & Feasible Frontier



Simplex Optimization Results:

- **Objective:** Maximize Z = 4,500 X + 6,800 Y
- **Constraints:** Working Capital (5X + Y <= 500), Tech Hours (2X + 8Y <= 1200), Min Sales (X >= 40), Min Service (Y >= 60).
- **Optimal Solution (X*, Y*): 74 Hardware Units & 132 AMC Contracts**.
- **Max Monthly Margin: INR 1,226,304** (+28.4% uplift).
- **Dual Value:** Technician shadow wage = **INR 850/hr** (3.0x return on technician salary).

STRATEGIC RECOMMENDATIONS

Managerial Strategy & Implementation Roadmap

1. Align Sales Compensation:

- Transition from gross volume quotas to service-attached margin quotas.
- Require minimum 60% AMC attachment threshold for bonus eligibility.

2. Introduce Tiered POS Bundling:

- Bundle 'Silver AMC' into retail financing EMIs (adds < INR 250/mo).
- Pre-configure 'Gold & Platinum' AMCs for SME/Enterprise accounts.

3. Defend the 2.5-Day SLA:

- Expand mobile service vans to service high-density corporate corridors.
- Reinvest service margins into technician hiring (shadow return: 3.0x).

4. Automated Lifecycle Alerts:

- Trigger automated renewal nudges at 10-month mark to sustain >80% retention.